



Cambridge IGCSE™

COMBINED SCIENCE

0653/43

Paper 4 Theory (Extended)

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MARK SCHEME

Maximum Mark: 80

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **14** printed pages.

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: - the specific content defined in the mark scheme - the specific skills defined in the mark scheme or in the level descriptions - the standard of response required as exemplified by the standardisation scripts.
2	Examiners must award whole marks (not half marks, or other fractions).
3	Examiners must award marks positively . This means that: - marks are not deducted for errors or omissions - marks are awarded for correct/valid answers as defined in the mark scheme and also for other valid answers which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate - the quality of spelling, punctuation and grammar are judged only when the mark scheme indicates that these features are specifically assessed in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must consistently apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the full range of marks defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and must not award marks with grade thresholds or grade descriptions in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	correct point or mark awarded
	incorrect point or mark not awarded
	information missing or insufficient for credit
	incorrect or insufficient point ignored while marking the rest of the response
	contradiction in response, mark not awarded
	benefit of the doubt given
	error carried forward applied
	response is too vague or there is insufficient detail in response
	incorrect point or mark not awarded

Annotation	Meaning
	point has been noted, but no credit has been given or blank page seen
	key point attempted / working towards marking point / incomplete answer / response seen but not credited / blank page seen
	blank page
	unclear response
	two statements are linked
	follow through
	irrelevant material that does not answer the question

Question	Answer	Marks	Guidance												
1(a)	<table><tr><th>name of part</th><th>letter</th><th>function of part</th></tr><tr><td>stigma</td><td>P</td><td>pollination</td></tr><tr><td>filament</td><td>S</td><td>holds up anther</td></tr><tr><td>petal</td><td>T</td><td>attracts insects</td></tr></table> ... one mark each row	name of part	letter	function of part	stigma	P	pollination	filament	S	holds up anther	petal	T	attracts insects	3	ALLOW supports / AW ALLOW attracts pollinators IGNORE attracts animals
name of part	letter	function of part													
stigma	P	pollination													
filament	S	holds up anther													
petal	T	attracts insects													
1(b)	pollen (grains) transferred (in pollination) ; fertilisation involves fusion of nuclei (of pollen and ovule) ;	2	ALLOW ovule is not used in pollination ALLOW fertilisation is the fusion of, gametes / ovule and pollen ALLOW contact between male and female gametes / pollen enters, reaches ovule / ovum IGNORE pollen reaches ovary												
1(c)	any three from: required for uptake of mineral ions ; lower concentration (of minerals ions) in soil ; movement against a concentration gradient ; uses / requires energy (from respiration) ;	3	ALLOW ATP as alternative wording (AW) for energy IGNORE references to water												

Question	Answer	Marks	Guidance
2(a)(i)	D ; C ;	2	
2(a)(ii)	X (valve) / it ensures one-way flow (of blood) / prevents backflow (of blood) ; Y (muscle) it contracts to push / pump (blood) ;	2	IGNORE just, one-way valve ALLOW compresses / squeezes, to push... IGNORE just, it contracts

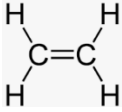
Question	Answer	Marks	Guidance
2(b)(i)	144 and 68 (seen) or 76 ; $(76 \times 100) \div 68$; 112(%) ;	3	ALLOW 112 no working = 3 marks ALLOW $\frac{144}{68} \times 100 - 100$ ALLOW 111.7647... / 110% (2 s.f.) ALLOW max 2 for incorrect readings but correct method (1) and evaluation (1)
2(b)(ii)	<i>any three from:</i> less muscle contraction ; (muscles) require less energy ; (muscles) require less oxygen ; decrease in respiration ;	3	mention of decrease / less only required once IGNORE heart contraction ALLOW less oxygen needed by the body ALLOW more energy needed by the body IGNORE provide oxygen

Question	Answer	Marks	Guidance
3(a)(i)	as nitrogen ;	1	ALLOW sulfur / S / N
3(a)(ii)	lipase acts on / digests / is specific to, fats / oils / lipids ; active site is not complementary (to proteins / substrate) ; no enzyme-substrate complexes form (with proteins) ;	3	ALLOW protease digests proteins ALLOW as AW complementary: does not fit protein / wrong shape for protein / can't bind with protein
3(b)(i)	herbivore ;	1	more than one circled = 0 marks
3(b)(ii)	Sun ; chlorophyll ; chemical ; <u>inchworm</u> ;	4	ALLOW sunlight

Question	Answer	Marks	Guidance															
4(a)(i)	$2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$; ; M1 correct formulae M2 balancing, dependent on correct formula	2	IGNORE state symbols ALLOW multiples / fractional balancing ALLOW max 1 if only mistake is 2Cl															
4(a)(ii)	(pale) yellow – green ;	1	ALLOW green NOT yellow															
4(a)(iii)	<table><thead><tr><th>statement</th><th>true</th><th>false</th></tr></thead><tbody><tr><td>Chlorine contains covalent bonds.</td><td>✓</td><td></td></tr><tr><td>Chlorine is less reactive than bromine.</td><td></td><td>✓</td></tr><tr><td>Chlorine displaces iodine from aqueous potassium iodide.</td><td>✓</td><td></td></tr><tr><td>Chlorine bleaches damp litmus paper.</td><td>✓</td><td></td></tr></tbody></table> ;;;	statement	true	false	Chlorine contains covalent bonds.	✓		Chlorine is less reactive than bromine.		✓	Chlorine displaces iodine from aqueous potassium iodide.	✓		Chlorine bleaches damp litmus paper.	✓		3	4 correct = 3 marks 2–3 correct = 2 marks 1 correct = 1 mark
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Chlorine displaces iodine from aqueous potassium iodide.	✓																	
Chlorine bleaches damp litmus paper.	✓																	
4(b)	8 cross electrons on outer shell of sodium ion ; 7 dot electrons on outer shell of the chloride ion and 1 cross electron ; (1)+ charge on sodium ion and (1) – charge on chloride ion ;	3	ALLOW all dots / all crosses / any combination provided the number of e is correct IGNORE inner electrons															

Question	Answer			Marks	Guidance
5(a)(i)	name	formula	atoms in the formula	2	ALLOW elements in any order
	<i>potassium hydroxide</i>	<i>KOH</i>	<i>1 potassium 1 oxygen 1 hydrogen</i>		
	<i>ammonium hydroxide</i>	<i>NH₄OH</i>	<i>1 nitrogen 5 hydrogens 1 oxygen ;</i>		
	<i>sodium hydrogencarbonate</i>	<i>NaHCO₃ ;</i>	<i>1 sodium 1 hydrogen 1 carbon 3 oxygen</i>		
5(a)(ii)	some bases are not soluble (in water) ;			1	ALLOW an alkali is a soluble base
5(b)(i)	salt + water ;			1	either order
5(b)(ii)	increasing temperature increases the kinetic energy of the particles / particles move faster ; frequency of (successful) collisions increases / the chance of (successful) collisions increases ; more collisions have the activation energy for a reaction ;			3	ALLOW reactants as AW for particles

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Question	Answer	Marks	Guidance
6(a)	M1 refinery gas at left end and bitumen at right end ; M2 correct order for naphtha and diesel oil ;	2	ALLOW 1 mark for any two in the right order / correct order reversed refinery gas – naphtha – diesel oil – bitumen = 2
6(b)(i)	C ₁₀ H ₂₂ ;	1	ALLOW H ₂₂ C ₁₀
6(b)(ii)	alkane ;	1	
6(b)(iii)	M1 conditions: high temperature and <u>catalyst</u> ; M2 for a double bond / indication that an alkene is formed ; M3 for correct number of hydrogens on each carbon ;	3	ALLOW 450–700°C AW high temperature ALLOW high pressure or 70 atm and catalyst or zeolite  ALLOW MP3 for C ₂ H ₄
6(c)(i)	sulfur dioxide ;	1	ALLOW oxides of nitrogen / sulfur oxide IGNORE water / SO
6(c)(ii)	<i>any two from:</i> planting trees / plants ; decreasing use of fossil fuels ; increasing use of renewable energy ; AVP ;	2	IGNORE reduction in livestock farming ALLOW named renewable energy AVP e.g. qualified lifestyle changes / carbon capture / use public transport

Question	Answer	Marks	Guidance
7(a)(i)	microwave ;	1	
7(a)(ii)	transverse: water / (named) electromagnetic / S – waves ; longitudinal: sound / P – waves ;	2	ALLOW any named electromagnetic wave / ocean – wave / rope – wave
7(b)(i)	$(v =) 2\pi r \div T$ =in any form (numerical or algebraic) e.g. $2\pi \times 3.85 \times 10^5 \div 27.3 \times 24$ OR $= 2\pi \times 3.85 \times 10^5 \div 655.2$; 3690 (km / h) ;	2	ALLOW 3692 / 3700 / 3.7×10^3
7(b)(ii)	(speed =) distance $\div$ time in any form (numerical or algebraic) e.g. $3.85 \times 10^8 \times 2 \div 3.0 \times 10^8$; 2.6 (s) ;	2	ALLOW $3.85 \times 10^5 \div 3.0 \times 10^n$ or 1.28×10^n or 2.6×10^n or their distance $\div$ their light – speed ALLOW answers that round to 2.6 at two s.f.

Question	Answer	Marks	Guidance
8(a)(i)	$W = mg = 6\,500 \times 9.8 = 63\,700$; (= 64 000 N)	1	ALLOW shows that the ratio of weight $\div$ mass = 9.8 (at two s.f.)
8(a)(ii)	M1 total area: 2×0.65 OR 1.30 seen (m^2) ; M2 use of $p = \frac{F}{A}$ OR $64\,000 \div 1.30$; M3 49 000 ; Pa or N / m^2 ;	4	NOT M2 or M3 if use of mass = 6 500 ALLOW answers that round to 49 000 at 2 s.f. IGNORE 50 000 (1 s.f. on answer line) ALLOW 2 marks for 98 000 (one foot) ALLOW Pascals / pascals / pa / 49 kPa / 49 kN / m^2

Question	Answer	Marks	Guidance
8(b)	M1 Use of KE equation to find final speed: e.g. $52\,000 = \frac{1}{2} 6\,500 \times v^2$ OR $v = \sqrt{(2 \times 52\,000 \div 6\,500)}$ OR $\frac{1}{2} mv^2$; M2 final speed = 4.0 (m / s) ; M3 change in speed (= 4.0 – 1.0 =) 3.0 (m / s) ; M4 <i>evidence of</i> $a = \frac{\Delta v}{\Delta t}$ / 3.0 ÷ 15 ; M5 0.20 (m / s²) ;	5	ALLOW candidate substitutes $v = a \times t$ into KE equation and evaluates consistently (= 2 marks)
9(a)	M1 heating element particles vibrate / gain KE / move faster ; M2 collisions transfer energy (to water particles) / energy is transferred from particle to particle ;	2	IGNORE refs to electrons / convection / water particles gain KE in M1

Question	Answer	Marks	Guidance
9(b)(i)	use of $E = Pt$ / $2.2 \times 5.0 \times 60$; 660 kJ ; efficiency = $\frac{\text{useful (energy) output}}{\text{total (energy) input}} \times 100\%$ OR $\frac{610}{660} \times 100\%$; 92(%) ;	4	ALLOW 92.4 ALLOW 3 marks for 0.92
9(b)(ii)	one heater still works if other breaks ORA / both heaters receive full supply p.d. ;	1	ALLOW lower (total) resistance / more current (in circuit) / greater heating effect / can be operated independently

Question	Answer	Marks	Guidance
9(b)(iii)	M1 use of $P=IV$ to find total current $I = 2\,200 / 230 = : 9.6 \text{ A}$; M2 calculation of current in heater $B = 9.6 - 4.2 = : 5.4 \text{ A}$; M3 use of $R=V/I = 230 \div 5.4 = : 43 (\Omega)$;	3	ALLOW 42.593... ALLOW for 1 mark, $230 \div$ their current OR their voltage $\div$ their current